Student Performance Anomaly Detection

Interactive demo using **Isolation Forest** to detect anomalous student profiles based on behavioral and academic attributes.

Dataset Overview

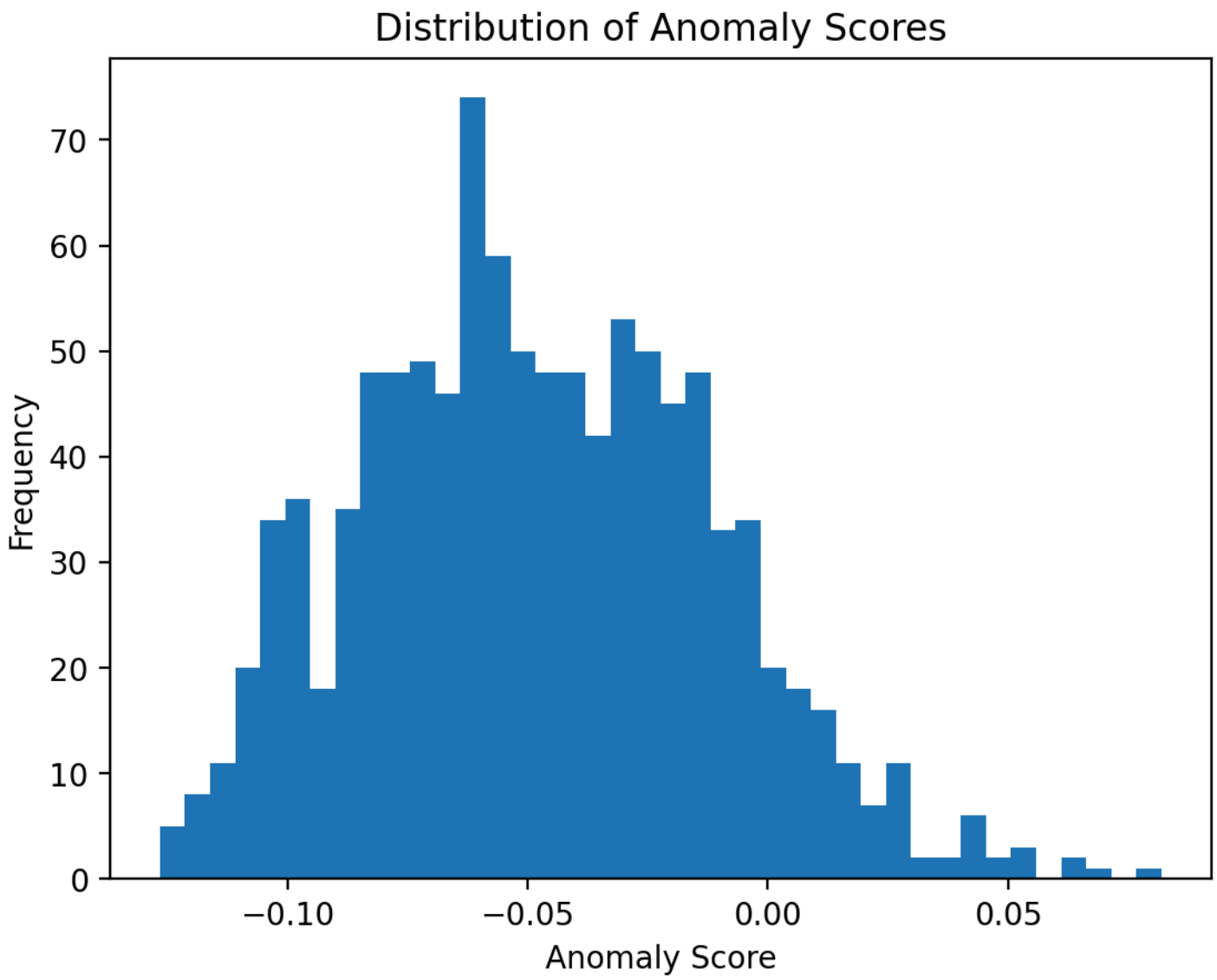
Total Students
1044

Features
32

Model
Isolation Forest

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	reason	guardian	traveltime	studytime	failures	schoolsup	famsup
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	course	mother	2	2	0	yes	no
1	GP	F	17	U	GT3	T	1	1	at_home	other	course	father	1	2	0	no	yes
2	GP	F	15	U	LE3	T	1	1	at_home	other	other	mother	1	2	3	yes	no
3	GP	F	15	U	GT3	T	4	2	health	services	home	mother	1	3	0	no	yes
4	GP	F	16	U	GT3	T	3	3	other	other	home	father	1	2	0	no	yes

Anomaly Score Distribution



Most Anomalous Students

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	reason	guardian	traveltime	studytime	failures	schoolsup	fams
1005	MS	F	19	R	GT3	A	1	1	at_home	at_home	course	other	2	2	3	no	yes
390	MS	M	20	U	LE3	A	2	2	services	services	course	other	1	2	2	no	yes
882	MS	F	18	R	LE3	A	3	2	other	other	course	other	2	3	2	no	yes
980	MS	F	17	R	LE3	A	3	1	other	at_home	course	other	2	3	0	no	yes
674	GP	M	22	U	GT3	T	3	1	services	services	other	mother	1	1	3	no	no
247	GP	M	22	U	GT3	T	3	1	services	services	other	mother	1	1	3	no	no
829	MS	F	16	U	LE3	A	2	2	at_home	other	reputation	mother	2	4	0	no	no
336	GP	F	19	R	GT3	A	3	1	services	at_home	home	other	1	3	1	no	no
808	GP	M	21	R	LE3	T	1	1	at_home	other	course	other	2	2	2	no	yes
1000	MS	F	19	U	GT3	T	1	1	at_home	services	other	father	2	1	1	no	no

Interactive Student Check

Modify key attributes and detect whether the student profile is anomalous.

Age

18

Study time

2

Past failures

1

Absences

5

Going out frequency

3

Free time

3

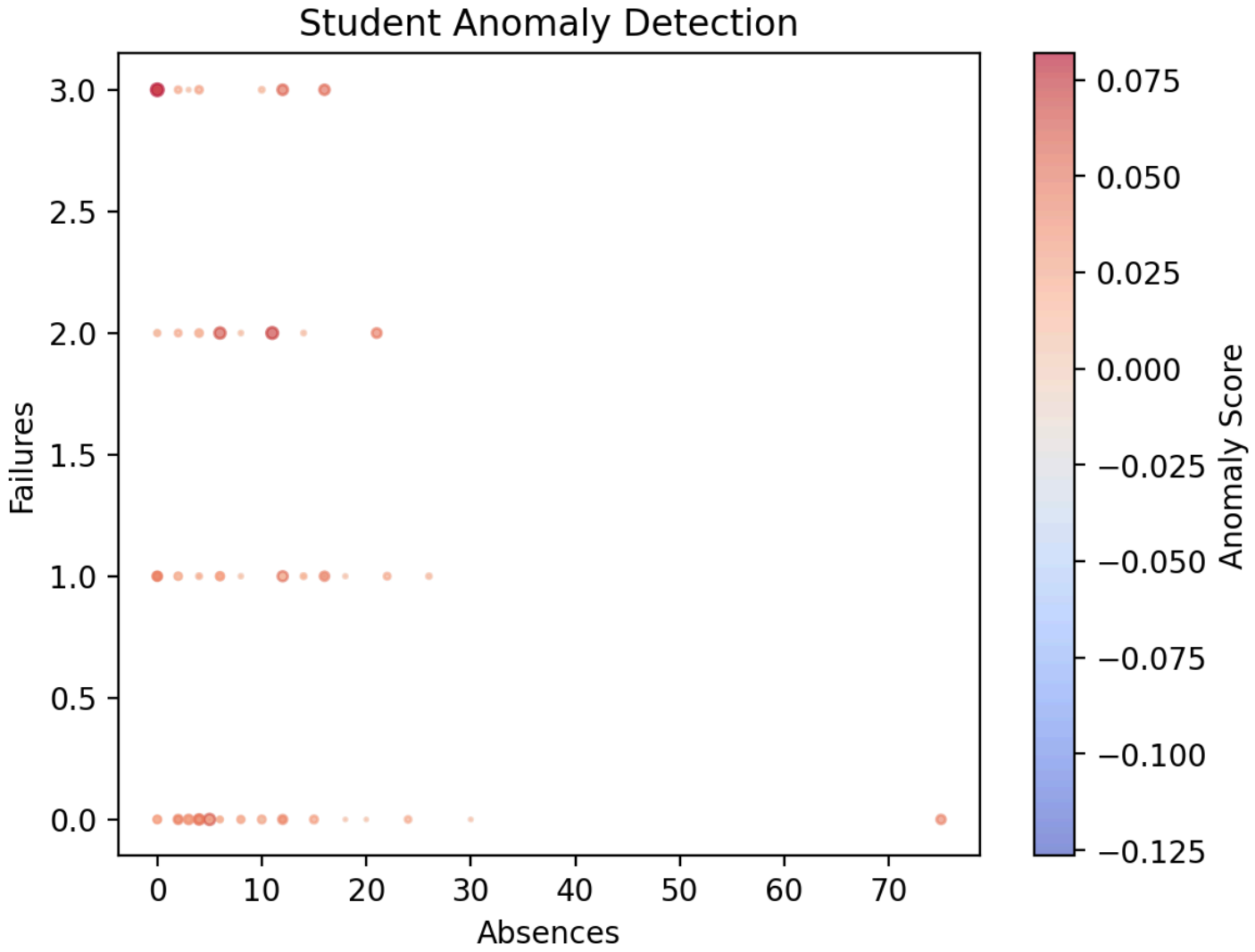
Detect anomaly

Prediction Result

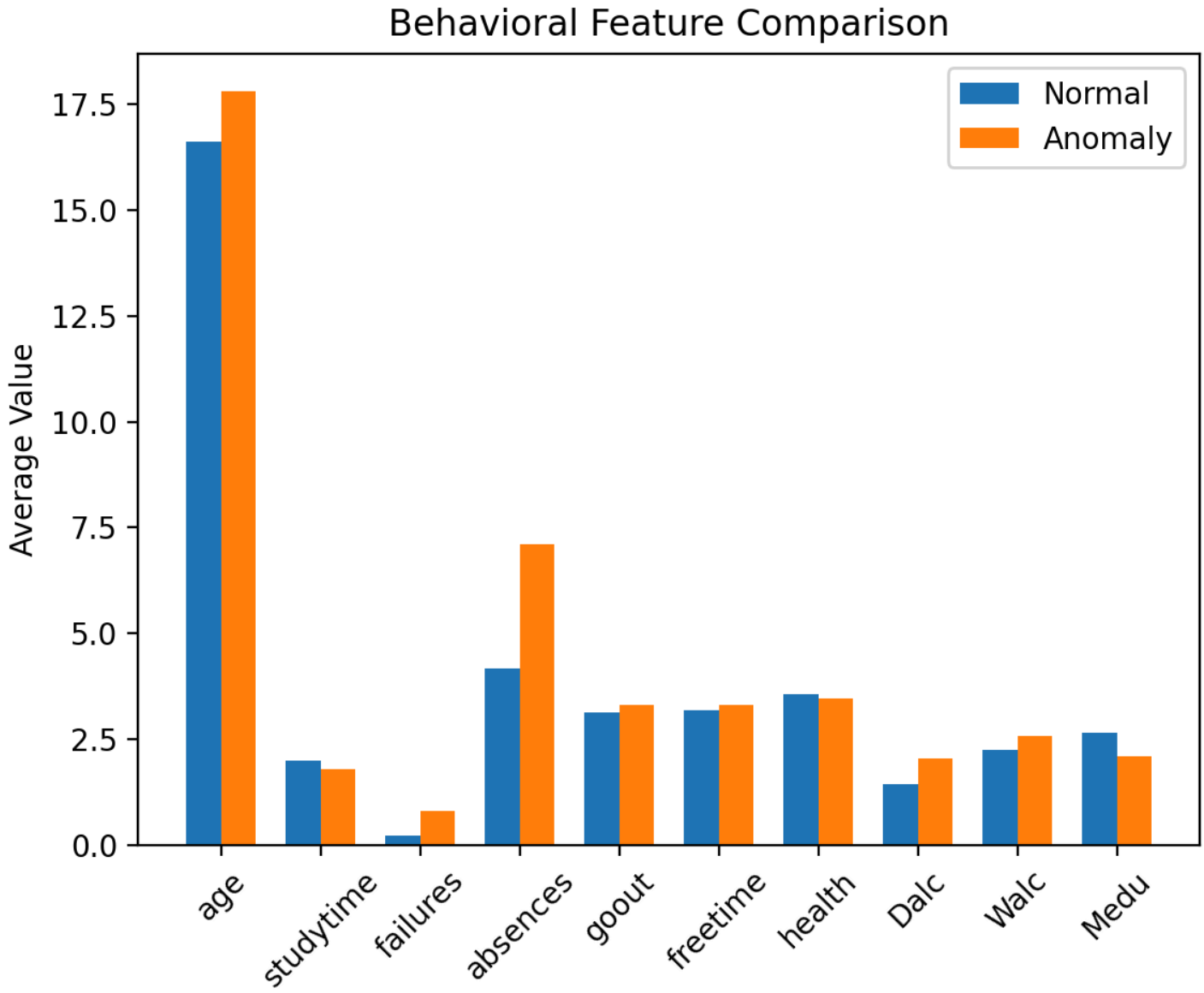
Anomaly score: `0.005`

This student profile appears anomalous.

Absences vs Failures (Anomaly Visualization)



Behavior Pattern Comparison: Normal vs Anomaly


Behavioral Feature Comparison
Normal
Anomaly
Average Value
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
age
studytime
failures
absences
goout
freetime
health
Dalc
Walc
Medu